Scenario-Based Exam Cheat Sheet

IERG4320/IEMS5726 Data Science in Practice | Answer Template + All Scenarios

Answer Template (3 Paragraphs)

情景题万能答题模板，按这三段写

P1: Objectives & Motivations (3%) – What + Why + Expected outcome. "The objective is to develop a [system] that [does what] using [technique]. Motivated by [R1], [R2], [R3]."

P2: Challenges & Limitations (2%) – 2–3 challenges + 1–2 limitations. "Key challenges: [C1], [C2], [C3]. Limitations: [L1], [L2]."

P3: Data Collection & Format (3%) – Source (API/sensor/DB/web) + format (tabular/image/text/audio/time series) + features/dimensions. "Data from [source]. Format: [type] with [F1, F2, ...]. Size: [N]×[M]."

Common Building Blocks

通用答题素材：挑着用

Common Challenges (pick 2–3)

Data quality (missing/noise/outliers) 数据质量 | Class imbalance (use F1/MCC) 类别不平衡 | Cold start 冷启动 | Overfitting (dropout, CV) 过拟合 | High dimensionality (PCA) 高维 | Real-time requirement 实时 | Computational cost (GPU) 计算成本 | Privacy/ethics 隐私伦理 | Domain knowledge 领域知识 | Black swan 黑天鹅

Data Sources & Formats

Sources: API (yfinance, Google Cloud) | DB (SQL, MongoDB) | Web scraping | Public (Kaggle, IMDB, ImageNet) | Sensors (CCTV, mic, IoT) | User logs (clicks, ratings).

Formats: Tabular (CSV, rows×cols) | Time series (OHLCV) | Text (tokens, TF-IDF, embeddings) | Image (H×W×C) | Audio (waveform 1D, spectrogram 2D) | Graph (adjacency + features).

Metrics

Classify: Acc, Prec, Recall, F1, MCC, AUC | Regress: R^2, MAE, RMSE | Cluster: Silhouette↑, CH↑, DB↓ | NLP: BLEU, METEOR, CIDEr, SPICE | ImgGen: FID↓ | Speech: MOS↑ | 3D: Chamfer↓, F-Score↑ | Finance: Sharpe↑, Drawdown↓ | RL: win rate

Scenario Quick Cards

每个场景的要点速查

A. Prediction & Forecasting

A1. Lottery / Mark Six Forecast 彩票预测

O: Predict next draw using historical data, probability, regression.

C: Random by nature; no guaranteed pattern; overfitting to noise; small sample relative to combination space (C_{49}^{6}=13.98M).

D: Historical draw results from official website. Tabular: draw_date, 6 winning numbers. Time series format.

A2. Stock Price Prediction 股票预测

O: Predict future price/trend using LSTM on historical OHLCV data.

C: Black swans, market psychology, reflexivity, overfitting, non-stationary data.

D: yfinance API. Time series: Open, High, Low, Close, Volume. Sliding window (e.g. 30 days) as input.

A3. Automated Trading Agent 自动交易

O: DQN agent learns buy/sell/hold strategy to maximize portfolio return.

C: Sparse reward, large state space, market regime changes, transaction costs.

D: yfinance API. State: prices + portfolio. Actions: buy/sell/hold. Reward: profit/loss.

B. Recommender Systems

B1. Movie/Music Recommendation 电影/音乐推荐

O: Recommend items based on user preferences (Netflix/Spotify).

C: Cold start (new users/items), data sparsity, scalability, over-specialization.

D: User-item rating matrix (sparse). User profiles, item metadata (genre, tags). Interaction logs (clicks, plays, ratings).

B2. E-commerce Product Recommendation 电商推荐

O: Recommend products to increase conversion rate (Amazon).

C: Same as B1 + fast-changing inventory, session-based behavior.

D: Purchase history, browsing logs, product features. Tabular + text (reviews).

C. NLP Scenarios

C1. Fake News Detection 假新闻检测

O: Classify news articles as real or fake using NLP models.

C: Polysemy, context-dependent meaning, evolving language, adversarial manipulation.

D: News articles with labels. Text data. Tokenized → embeddings. Classic: TF-IDF (25k×5k). DL: BERT tokenizer (max_length=256).

C2. Sentiment Analysis (Movie Reviews) 情感分析

O: Classify reviews as positive/negative (IMDB dataset).

C: Sarcasm, negation, context dependency, word order matters.

D: IMDB: 25k train + 25k test. Text → tokenized → embeddings. Labels: pos/neg.

C3. Chatbot / Assistant 聊天机器人

O: Conversational AI for customer service using GPT/LLaMA.

C: Intent understanding, multi-turn context, hallucination.

D: Conversation logs, FAQ pairs (query, response). Fine-tune GPT/LLaMA.

C4. Translation / Summarization 翻译/摘要

O: Translate or summarize text using seq2seq Transformer (T5).

C: Word order, idioms, coherence, hallucination, rare languages.

D: Parallel corpus or document corpus. Text → tokenized. Markov chain as non-DL baseline.

D. Audio / Speech

D1. Vocal Separation (Karaoke) 人声分离

O: Isolate vocals from music for karaoke or remixing.

C: Frequency overlap between voice and instruments, harmonic complexity, temporal masking.

D: Audio files (WAV/MP3). Waveform (1D) or spectrogram (2D). Use librosa or Demucs.

D2. Text-to-Speech (TTS) 语音合成

O: Convert text to natural-sounding speech (online dictionary, movie effects).

C: Natural prosody, emotion expression, speaker identity, robotic sound.

D: Text + paired audio recordings. Text → mel-spectrogram

→ waveform. Tacotron2 + WaveNet. Evaluate with MOS.

D3. Speech Recognition (STT) 语音识别
O: Convert speech to text (smart home, voice assistant).
C: Accents, background noise, speaker variation, homophones.
D: Audio recordings + transcriptions. Waveform → features. Use HuBERT / Wav2Vec 2.0. Speaker-dependent vs independent.

E. Computer Vision

E1. Accident / Anomaly Detection (CCTV) 事故/异常检测
O: Detect falls, choking, violence in elderly homes / public areas via CCTV.
C: Real-time requirement, occlusion, lighting changes, false positives, privacy.
D: CCTV video frames. Image sequences (H×W×3). CNN for detection (ResNet), action recognition. Labels: normal/accident.

E2. Medical Image Diagnosis 医疗影像诊断
O: Detect diseases (fractures, tumors) from X-ray/CT images.
C: Small dataset, class imbalance (rare diseases), need expert labels, interpretability required.
D: Hospital image database. Images (DICOM format → pixel arrays). Labels from radiologists. Use pre-trained CNN + fine-tune.

E3. Traffic Surveillance 交通监控
O: Detect vehicles, accidents, violations from cameras.
C: Real-time, weather/lighting, occlusion, diverse vehicle types.
D: Camera feeds. Video frames. Object detection (Mask R-CNN). Labels: vehicle/violation type.

E4. Image Captioning 图像描述
O: Generate text for images (accessibility for visually impaired).
C: Identify objects+actions+relationships+grammar. Feature bottleneck.
D: Image-caption pairs (COCO). 224×224×3 + text. CNN+RNN or ViT+Transformer. Eval: BLEU, CIDEr.

F. Generation

F1. AI Art / Marketing Image Generation AI艺术/营销图生成
O: Generate images from text descriptions for design/advertising.
C: High dimensionality (pixels), must imagine background, copyright concerns, quality control.
D: Text-image pairs. Text prompt → DALL-E/Stable Diffusion. Evaluate with FID. CLIP ranks outputs.

F2. DeepFake Detection 深度伪造检测
O: Detect manipulated videos to prevent fraud/misinformation.
C: Increasingly realistic fakes, best AI only 80% detection, adversarial attacks.
D: Real + fake video pairs. Video frames (face crops). Binary classification. Features: blinking, lip sync, skin texture.

F3. 2D-to-3D Reconstruction 2D转3D
O: Generate 3D model from a single 2D image (game assets, architecture).
C: Infinite 3D shapes can explain one 2D view; needs strong priors; floater artifacts.
D: RGB images + 3D ground truth (Objaverse). Image → TripoSR (DINOv1 + Transformer + NeRF → mesh). Eval: Chamfer Distance, F-Score.

F4. Text-to-3D Generation 文字转3D
O: Generate 3D models from text descriptions (Shap-E).
C: Lower fidelity than manual modeling, topological errors, floaters, bottlenecked by 2D model.
D: Text-3D pairs. Text prompt → image diffusion → point cloud diffusion → mesh. ∼13s per asset.

G. Game & RL

G1. Card Game AI (Blackjack) 21点AI
O: Train agent to play Blackjack optimally using RL.
C: Sparse reward (only at end), stochastic environment, exploration vs exploitation.
D: Simulated games. State: (player_sum, dealer_card, usable_ace). Actions: hit/stand. Use D3QN, A3C, or AlphaZero+MCTS.

G2. Board Game AI (Chess/Go) 棋类AI
O: Master board games via self-play (AlphaZero).
C: Huge state space, long-term planning, MCTS cost.
D: Self-play games. State: board position. MCTS + dual-headed net (policy+value).

G3. Multi-Game Agent 多游戏智能体
O: Single agent plays multiple games without forgetting.
C: Catastrophic forgetting: new task overwrites old weights.
D: Multiple game envs. EWC protects important weights. Train A → B(+EWC) → C.

H. Robot Learning

H1. Robot Locomotion 机器人行走
O: Train humanoid to walk using RL + transfer learning.
C: High-dim continuous actions, balance, sim-to-real gap.
D: Simulated env. State: joint angles. Transfer: crawl → wall-walk → free-walk.

H2. Robot Manipulation 机器人抓取
O: Train robot arm to grasp objects.
C: Object variety, precision, deformable objects.
D: Camera + joint data. Imitation / inverse RL / diffusion policies.

Which Model for Which Problem?

问题类型→模型速查

Problem	Models
Classification	DT, RF, SVM, KNN, ANN/MLP
Regression	Linear, Ridge, LASSO, SVR, ANN
Clustering	K-Means, Agglomerative, Spectral
Recommendation	CF, Content, Hybrid, MF/SVD, Wide&Deep, GCN
Time series	EMA, LSTM, RNN
Text classify	TF-IDF+NB, NBoW, BERT, TinyBERT
Text generation	Markov, GPT, LLaMA
Speech (TTS)	WaveNet, Tacotron2
Speech (STT)	Wav2Vec 2.0, HuBERT
Image classify	AlexNet, VGG, ResNet, ViT
Image caption	CNN+RNN+Attn, ViT+Transformer
Image generation	GAN, DALL-E, Diffusion
3D generation	TripoSR, Shap-E
Game agent	Q-Learning, DQN, D3QN, A3C, AlphaZero
Robot control	RL, imitation, behavior cloning, inverse RL